

Evolution

THE ORGANISMS MOST SUITED TO SURVIVE IN AN ENVIRONMENT ARE MOST LIKELY TO PASS ON THEIR GENES.

SEE ALSO

◀ 20–21 Variety of life

◀ 42–43 Reproduction I

Adaptations

82–83 ▶

Genetics I

84–85 ▶

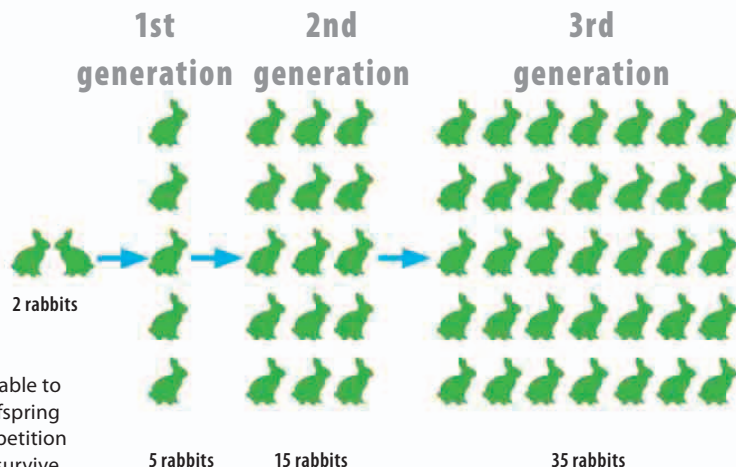
Set out by English naturalist Charles Darwin in 1859, the theory of evolution by natural selection was one of the most controversial scientific theories ever. It has become accepted since, and has been updated to include the role of genes.

The drive to breed

Everything an organism does is meant to increase the chance of it producing as many surviving offspring as possible. These offspring compete with each other and other species for limited resources, such as food, water, and a place to live. Those best able to survive are the ones that pass on their genes to the next generation. The individuals that cannot compete die without producing young, so their genes are not passed on.

▷ Increasing genes

Rabbits have a very high reproduction rate, with one female able to produce 70 young in just one year. The following year, her offspring could potentially produce almost 5,000 more. In reality, competition between all these rabbits is so fierce that far fewer than this survive.



Natural selection

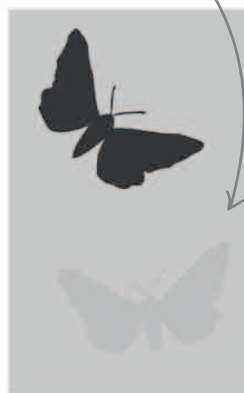
The most successful, or “fit,” offspring are the ones with genes that allow them to out-compete their rivals. When they mate, their fit genes are passed on to the next generation. This is called natural selection. Eventually every animal in the species has the fit genes—meaning the species gradually evolves over time.



△ Adding variety

Most variation between animals is the result of sexual reproduction. Every offspring inherits a slightly different mixture of genes from both parents. The variation in color of these moths ensures that at least some of the offspring will survive if the habitat begins to change.

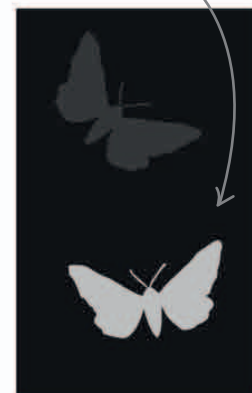
pale peppered moth is harder for a predator to see



△ Pale moths hidden

Before the Industrial Revolution, most peppered moths were pale and could hide in the lichens growing on tree trunks, while the darker moths stood out.

pale peppered moths are easier to see for predators, so experience a decline in numbers



△ Pale moths stand out

Then soot from factories killed the lichens, making tree trunks darker. The pale moths then became more preyed upon, which made the dark moths more common.